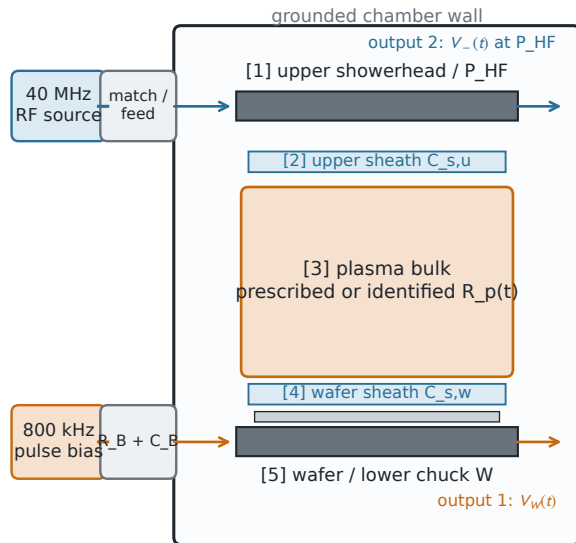


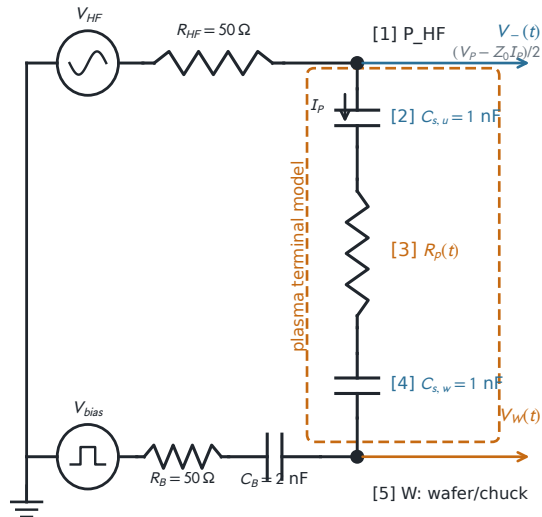
Dual-frequency CCP: apparatus and circuit use the same vertical order

Labels [1]-[5] map hardware to nodes and elements; V_W and V_- are distinct outputs.

(a) Chamber and named measurement planes



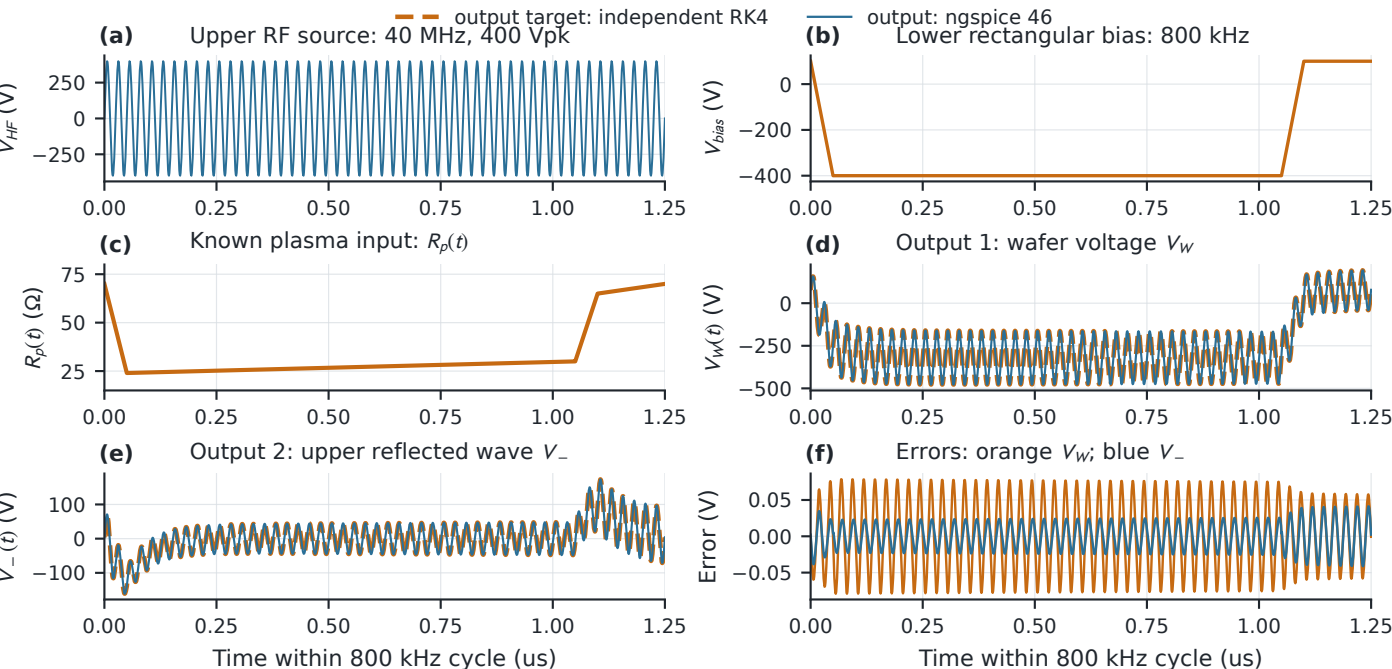
(b) Top-to-bottom ngspice equivalent circuit



Outputs: V_W = wafer/chuck to chamber ground; V_- = reflected wave at P_HF.
 $Z_0 = 50 \text{ ohm}$ and current is positive into the chamber; diagram not to scale.

Forward problem: known inputs produce the two intended device observables

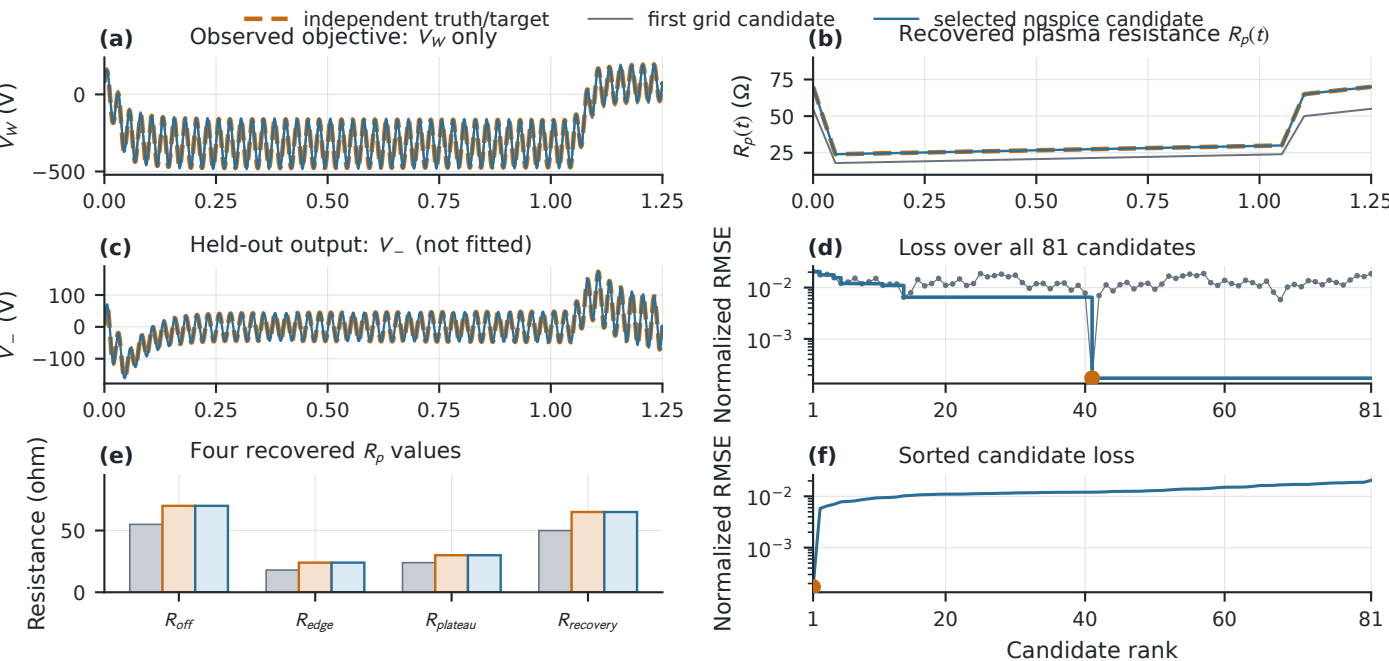
Source waveforms and $R_p(t)$ are inputs; wafer voltage and upper reflected voltage are outputs.



Final two bias cycles; 2501 target samples. Wafer RMSE=0.0527 V, nRMSE=1.717e-04.
Upper reflection RMSE=0.0195 V, nRMSE=3.989e-04.

Inverse problem A: wafer voltage alone recovers bounded $R_p(t)$

Only the named objective waveform is ranked; the other output is retained as a held-out circuit check.

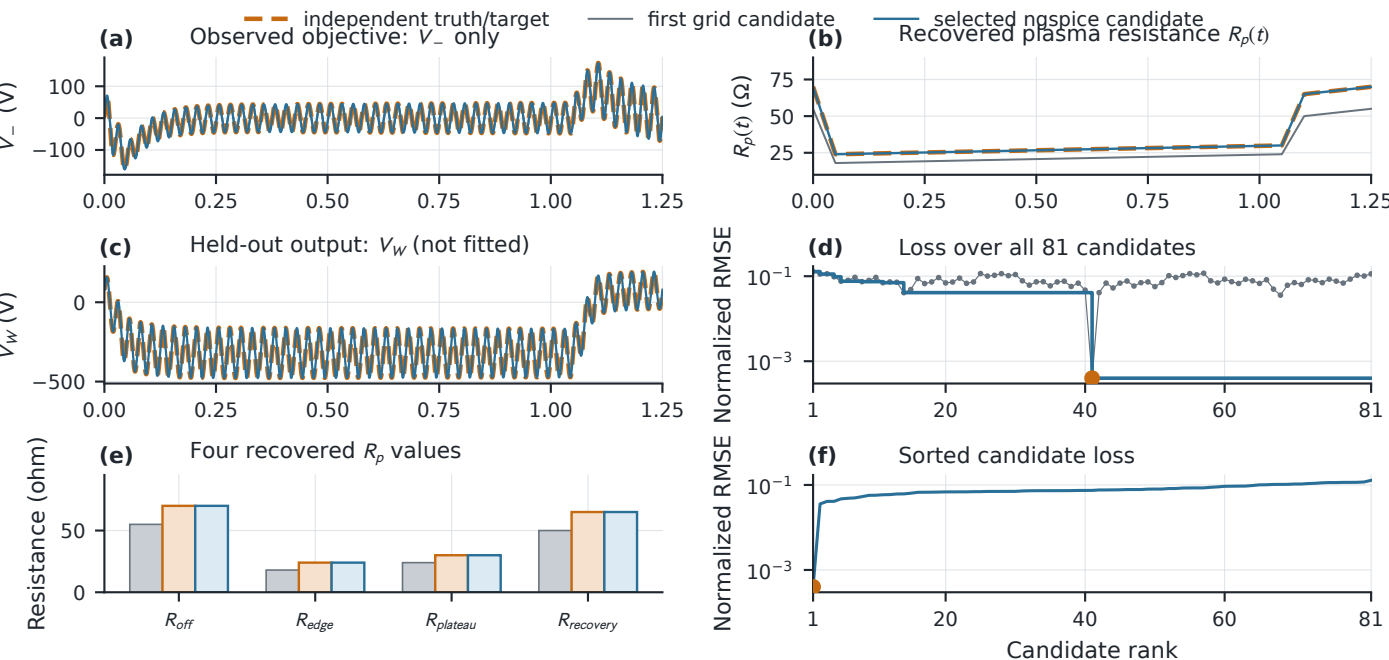


Selected values: off=70, edge=24, plateau=30, recovery=65 ohm.

Objective nRMSE=1.717e-04; held-out nRMSE=3.989e-04; profile RMSE=2.63e-11 ohm; 81/81 solves succeeded.

Inverse problem B: upper reflection alone recovers bounded $R_p(t)$

Only the named objective waveform is ranked; the other output is retained as a held-out circuit check.



Selected values: off=70, edge=24, plateau=30, recovery=65 ohm.

Objective nRMSE=3.989e-04; held-out nRMSE=1.717e-04; profile RMSE=2.63e-11 ohm; 81/81 solves succeeded.